## PINOUT TABLES - one table per connector, bus plug, splice and end of the set; every pin, the wire on it and where its other end lands

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w111_c | 20 | White | 🟦🟨🟦🟨 | Injector 4 drive (-) | DRB102 receptacle (RH firewall) pin 24 | 630 | ST185-AB-cabin | INJ4 |
| 2 | w110_c | 20 | White | 🟦🟨🟦🟦 | Injector 3 drive (-) | DRB102 receptacle (RH firewall) pin 23 | 630 | ST185-AB-cabin | INJ3 |
| 3 | w109_c | 20 | White | 🟦🟦⬜⬜ | Injector 2 drive (-) | DRB102 receptacle (RH firewall) pin 22 | 630 | ST185-AB-cabin | INJ2 |
| 4 | w108_c | 20 | White | 🟦🟦⬜🟨 | Injector 1 drive (-) | DRB102 receptacle (RH firewall) pin 21 | 630 | ST185-AB-cabin | INJ1 |
| 5 | w_a5_pin | 18 | Red | 🟦 | EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak... | ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode) | 310 | ST185-AB-cabin | SW12 |
| 6 | w19 | 20 | Orange | ⬜🟦⬜ | +8 V sensor supply | ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6) | 310 | ST185-AB-cabin | 8V |
| 7 | w_drain_ecu_a | 20 | Black | 🟨⬜🟦 | Shield drain | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 200 | ST185-AB-cabin | SHLD |
| 8 | w200_c | 20 | White | 🟦🟦⬜🟦 | Crank sensor signal | DRB102 receptacle (RH firewall) pin 74 | 630 | ST185-AB-cabin | CRANKC |
| 9 | w_trig2_pin | 20 | White | 🟦🟦🟨 | Cam sensor signal | At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V | 310 | ST185-AB-cabin | CAM |
| 10 | w123_c | 20 | White | ⬜⬜⬜ | Coil 4 trigger (IGT) | DRB102 receptacle (RH firewall) pin 12 | 630 | ST185-AB-cabin | COP4 |
| 11 | w122_c | 20 | White | ⬜⬜🟨 | Coil 3 trigger (IGT) | DRB102 receptacle (RH firewall) pin 11 | 630 | ST185-AB-cabin | COP3 |
| 12 | w121_c | 20 | White | ⬜🟨⬜ | Coil 2 trigger (IGT) | DRB102 receptacle (RH firewall) pin 10 | 630 | ST185-AB-cabin | COP2 |
| 13 | w120_c | 20 | White | ⬜🟨🟨 | Coil 1 trigger (IGT) | DRB102 receptacle (RH firewall) pin 9 | 630 | ST185-AB-cabin | COP1 |
| 14 | w38 | 20 | Blue | 🟨⬜⬜ | Pedal 1 signal main | accelerator pedal sensor pin 6 | 1090 | ST185-AB-cabin | APS |
| 15 | w201_c | 20 | White | 🟦🟦🟦🟦 | Coolant temp signal | DRB102 receptacle (RH firewall) pin 70 | 630 | ST185-AB-cabin | ECT |
| 16 | w202_c | 20 | White | 🟦🟨🟨🟨 | Intake temp signal | DRB102 receptacle (RH firewall) pin 33 | 630 | ST185-AB-cabin | IATC |
| 17 | w26_c | 20 | White | 🟦🟨⬜🟨 | MAP sensor signal | DRB102 receptacle (RH firewall) pin 17 | 630 | ST185-AB-cabin | MAPC |
| 18 | w_ac_kill | 20 | White | 🟦🟨🟦⬜ | A/C cancel (ACT) | OEM A/C amplifier pin 11 | 820 | ST185-AB-cabin | ACKILL |
| 19 | w_fp85 | 18 | White | ⬜⬜🟦 | Fuel pump relay (-) | Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU si... | 3130 | ST185-AB-cabin | FP |
| 20 | w81 | 20 | White | ⬜⬜ | ETB relay coil (-) | ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85); coil+ ... | 680 | ST185-CabinPower | ETB |
| 21 | w103_c | 20 | White | 🟦🟦🟨⬜ | Boost solenoid (-) | DRB102 receptacle (RH firewall) pin 58 | 630 | ST185-AB-cabin | BOOST |
| 22 | w128_c | 20 | White | 🟦🟦⬜ | Throttle pos 1 main | DRB102 receptacle (RH firewall) pin 42 | 630 | ST185-AB-cabin | ETBS |
| 23 | cab_fspur_l | 20 | Blue | ⬜🟦🟨 | FL conditioned out | to front VR conditioner pin 3 | 530 | ST185-AB-cabin | WSSF |
| 24 | w68 | 18 | Blue | 🟨🟨 | Sensor ground | DT06-12 plug (ECU splice point) pin 1 | 310 | ST185-AB-cabin | SGND |
| 25 | w70 | 18 | Black | 🟨🟨⬜ | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |
| 26 | w_str85 | 18 | White | ⬜🟨 | Start relay coil (-) | Start Relay (seated in the POWER box R3) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) s... | 680 | ST185-AB-cabin | STR |
| 27 | w_mrs_c | 20 | White | 🟦🟨⬜⬜ | EHPS pump speed | DRB102 receptacle (RH firewall) pin 84 | 630 | ST185-AB-cabin | MRSSPD |
| 28 | w76 | 18 | White | 🟦🟨 | EFI relay coil (-) | EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (... | 680 | ST185-CabinPower | EFI |
| 29 | w_fan85 | 18 | White | 🟨⬜ | Rad fan relay (-) | Radiator Fan Relay (on/off) (seated in the POWER box R4) - ACTIVE-LOW - ECU sinks coil- ... | 680 | ST185-AB-cabin | FAN |
| 30 | w16_c | 20 | White | 🟨🟦⬜ | Turbo speed signal | DRB102 receptacle (RH firewall) pin 53 | 630 | ST185-AB-cabin | TURBO |
| 31 | w48_c | 20 | White | 🟨🟨🟨 | Flex fuel signal | DRB102 receptacle (RH firewall) pin 31 | 630 | ST185-AB-cabin | FLEX |
| 32 | w67 | 18 | Orange | 🟦🟦 | +5 V sensor supply | DT06-12 plug (ECU splice point) pin 1 | 310 | ST185-AB-cabin | 5V |
| 33 | w129_c | 20 | White | 🟦🟦⬜ | Throttle pos 2 sub | DRB102 receptacle (RH firewall) pin 44 | 630 | ST185-AB-cabin | ETBS |
| 34 | w_a34 | 18 | Black | 🟨🟨⬜ | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |

### ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts TBD | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | spare, not used | | | | |
| 2 | | | | | spare, not used | | | | |
| 3 | | | | | spare, not used | | | | |
| 4 | | | | | spare, not used | | | | |
| 5 | w82_ac | 18 | Red | ⬜⬜ | ETB motor +12 V | ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85); coil+ ... | 680 | ST185-CabinPower | ETB |
| 6 | w203_c | 20 | White | 🟦🟦🟦🟦 | Oil temp signal | DRB102 receptacle (RH firewall) pin 61 | 630 | ST185-AB-cabin | OILT |
| 7 | w204_c | 20 | White | 🟦🟨🟨🟦 | Charge temp signal | DRB102 receptacle (RH firewall) pin 59 | 630 | ST185-AB-cabin | CIAT |
| 8 | | | | | spare, not used | | | | |
| 9 | w205_c | 20 | White | 🟦🟦🟦⬜ | Knock sensor signal | DRB102 receptacle (RH firewall) pin 71 | 630 | ST185-AB-cabin | KNOCK |
| 10 | w_acc_trig | 18 | White | 🟨🟦 | A/C clutch relay (-) | A/C Clutch Relay (on/off) (seated in the LOGIC box M5) - ACTIVE-LOW - ECU sinks coil- (85)... | 680 | ST185-AB-cabin | ACC |
| 11 | | | | | spare, not used | | | | |
| 12 | w_eps_trig | 18 | White | 🟦⬜ | EPS enable relay (-) | EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85); coil+... | 680 | ST185-AB-cabin | EPSEN |
| 13 | w_fan285 | 18 | White | ⬜🟦 | Cond fan relay (-) | Condenser Fan Relay (on/off) (seated in the POWER box R5) - ACTIVE-LOW - ECU sinks coi... | 680 | ST185-AB-cabin | FAN2 |
| 14 | | | | | spare, not used | | | | |
| 15 | w29_c | 20 | White | 🟦⬜🟦🟦 | Oil pressure signal | DRB102 receptacle (RH firewall) pin 28 | 630 | ST185-AB-cabin | OILPC |
| 16 | w32_c | 20 | White | 🟦🟦🟦🟨 | Fuel pressure signal | DRB102 receptacle (RH firewall) pin 14 | 630 | ST185-AB-cabin | FUELP |
| 17 | w_drain_ecu_b | 20 | Black | 🟨⬜🟦 | Shield drain B | 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17 | 200 | ST185-AB-cabin | SHLD |
| 18 | w124_c | 18 | White | 🟦🟨⬜ | ETB motor + | DRB102 receptacle (RH firewall) pin 45 | 630 | ST185-AB-cabin | ETBM |
| 19 | cab_rspur_r | 20 | Blue | 🟦🟦🟨🟨 | Rear right wheel spd | IX-WS-R inline to rear VRC (dash above ECU) pin 4 | 530 | ST185-AB-cabin | WSSR |
| 20 | cab_rspur_l | 20 | Green | 🟦🟦🟨🟨 | Rear left wheel spd | IX-WS-R inline to rear VRC (dash above ECU) pin 3 | 530 | ST185-AB-cabin | WSSR |
| 21 | cab_fspur_r_ac | 20 | White | ⬜🟦🟦 | FR conditioned out | to front VR conditioner pin 4 | 530 | ST185-AB-cabin | WSSF |
| 22 | w69 | 18 | Blue | 🟨🟨 | Sensor ground B | DT06-12 plug (ECU splice point) pin 1 | 310 | ST185-AB-cabin | SGND |
| 23 | w35_c | 20 | White | ⬜🟨🟦 | Coolant press signal | DRB102 receptacle (RH firewall) pin 26 | 630 | ST185-AB-cabin | CLNTP |
| 24 | w_fl_sig_pin | 20 | White | 🟦🟦🟦 | Fuel level signal | SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24 | 310 | ST185-AB-cabin | FLVL |
| 25 | w_b25_ac | 18 | Black | 🟨🟨⬜ | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |
| 26 | w215_c | 18 | White | 🟦🟦⬜ | ETB motor - | DRB102 receptacle (RH firewall) pin 46 | 630 | ST185-AB-cabin | ETBM |
| 27 | w_at_immob2 | 20 | White | 🟨🟨🟦 | Anti-theft input | Alarm module - main harness (DT06-12SA) pin 9 | 380 | ST185-AntiTheft | IMMOB |
| 28 | w_ign | 20 | Yellow | 🟦🟦🟦🟦 | Ignition sw sense | OEM J/B No.1 (LH kick panel) 1H pin 7, B-O pin 7 | 1100 | ST185-AB-cabin | IGNSW |
| 29 | w_vss_sig_c | 20 | White | 🟨🟨⬜ | Gearbox speed signal | DRB102 receptacle (RH firewall) pin 68 | 630 | ST185-AB-cabin | VSS |
| 30 | w_start | 20 | Yellow | 🟨🟨🟨 | Start request | LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R7... | 680 | ST185-AB-cabin | B30 |
| 31 | w_ac_btn | 20 | Yellow | 🟨🟨🟦 | A/C request button | SP-AC-BTN - A/C switch divider mid-point to ECU-B B31 | 310 | ST185-AB-cabin | ACBTN |
| 32 | w_evap_temp | 20 | White | ⬜🟦🟦 | Evap temp signal | SP-EVAP - evap thermistor + 2.2k pull-up to ECU-B B32 | 310 | ST185-AB-cabin | EVAP |
| 33 | w_aps_vpa2 | 20 | Blue | 🟨⬜⬜ | Pedal 2 signal sub | accelerator pedal sensor pin 3 | 1090 | ST185-AB-cabin | APS |
| 34 | w_b34_ac | 18 | Black | 🟨🟨⬜ | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |

### ECU-C - ECU Comms/Tuning Port

housing DTM06-6S + WM-6S | contacts 0462-201-2031 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_shd_can_trunk | 20 | Black | 🟨 | CAN screen trunk | CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen | 650 | ST185-CAN | CAN |
| 2 | | | | | spare, sealing plug | | | | |
| 3 | wc_h_ecu | 20 | White | 🟨 | CAN 1 high | CAN H join - ECU, CSB3, cluster, Pi, bulkhead | 250 | ST185-CAN | CAN |
| 4 | wc_l_ecu | 20 | Green | 🟨 | CAN 1 low | CAN L join - ECU, CSB3, cluster, Pi, bulkhead | 250 | ST185-CAN | CAN |
| 5 | | | | | spare, sealing plug | | | | |
| 6 | | | | | spare, sealing plug | | | | |

### BUS-5V - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w67 | 18 | Orange | 🟦🟦 | +5 V from ECU A32 | Link G4X plug A (centre dash) pin 32 | 310 | ST185-AB-cabin | 5V |
| 2 | w_rail_sp_5v_c | 20 | Orange | 🟦🟦 | +5 V throttle sensor | DRB102 receptacle (RH firewall) pin 41 | 520 | ST185-AB-cabin | 5V |
| 3 | w_5v_a_c39 | 20 | Orange | 🟦🟦 | +5 V MAP sensor | DRB102 receptacle (RH firewall) pin 16 | 520 | ST185-AB-cabin | 5V |
| 4 | w_5v_a_c40 | 20 | Orange | 🟦🟦 | +5 V turbo speed | DRB102 receptacle (RH firewall) pin 52 | 520 | ST185-AB-cabin | 5V |
| 5 | w_5v_brail_c45 | 20 | Orange | 🟦🟦 | +5 V oil pressure | DRB102 receptacle (RH firewall) pin 27 | 520 | ST185-AB-cabin | 5V |
| 6 | w_5v_brail_c46 | 20 | Orange | 🟦🟦 | +5 V fuel pressure | DRB102 receptacle (RH firewall) pin 13 | 520 | ST185-AB-cabin | 5V |
| 7 | w_5v_brail_c47 | 20 | Orange | 🟦🟦 | +5 V coolant press | DRB102 receptacle (RH firewall) pin 25 | 520 | ST185-AB-cabin | 5V |
| 8 | w_aps_vc1 | 20 | Red | 🟦🟦 | +5 V pedal sensor 1 | accelerator pedal sensor pin 4 | 1200 | ST185-AB-cabin | 5V |
| 9 | w210 | 20 | Red | 🟦🟦 | +5 V pedal sensor 2 | accelerator pedal sensor pin 1 | 1200 | ST185-AB-cabin | 5V |
| 10 | cab_fspur_5v | 20 | Red | 🟦🟦 | +5 V front VRC | to front VR conditioner pin 1 | 640 | ST185-AB-cabin | 5V |
| 11 | cab_rspur_5v | 20 | Red | 🟦🟦 | +5 V rear VRC | IX-WS-R inline to rear VRC (dash above ECU) pin 1 | 640 | ST185-AB-cabin | 5V |
| 12 | w_5v_bus_link | 20 | Orange | 🟦🟦 | +5 V to BUS-5V-2 | DT06-8 plug (ECU splice point) pin 1 | 200 | ST185-AB-cabin | 5V |

### BUS-5V-2 - DT06-8 plug (ECU splice point)

housing DT06-08SA + W8S; DT04-08PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_5v_bus_link | 20 | Orange | 🟦🟦 | +5 V from BUS-5V | DT06-12 plug (ECU splice point) pin 12 | 200 | ST185-AB-cabin | 5V |
| 2 | w_cru_5v_ac | 20 | Orange | 🟦🟦 | +5 V cruise ladder | resistor r_cruise on CSB3 pin 1 | 680 | ST185-AB-cabin | 5V |
| 3 | w_evap_5v | 20 | Orange | ⬜🟦🟦 | +5 V evap pull-up | 2.2k evap thermistor pull-up to +5V pin 2 | 200 | ST185-AB-cabin | EVAP |
| 4 | w44 | 20 | Orange | 🟦🟨🟦 | +5 V fuel lvl pullup | 470R fuel level pull-up to +5V pin 2 | 200 | ST185-AB-cabin | FLVL |
| 5 | | | | | spare, sealing plug | | | | |
| 6 | | | | | spare, sealing plug | | | | |
| 7 | | | | | spare, sealing plug | | | | |
| 8 | | | | | spare, sealing plug | | | | |

### BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w68 | 18 | Blue | 🟨🟨 | Sensor gnd from A24 | Link G4X plug A (centre dash) pin 24 | 310 | ST185-AB-cabin | SGND |
| 2 | w_rail_sp_gndout_c | 20 | Blue | 🟨🟨 | Gnd IAT/TPS/coolant | DRB102 receptacle (RH firewall) pin 34 | 520 | ST185-AB-cabin | SGND |
| 3 | w_gnd_a_c41 | 20 | Blue | 🟨🟨 | Gnd crank sensor | DRB102 receptacle (RH firewall) pin 75 | 520 | ST185-AB-cabin | SGND |
| 4 | w_gnd_a_c42 | 20 | Blue | 🟨🟨 | Gnd cam sensor | DRB102 receptacle (RH firewall) pin 65 | 520 | ST185-AB-cabin | SGND |
| 5 | w_gnd_a_c43 | 20 | Blue | 🟨🟨 | Gnd turbo speed | DRB102 receptacle (RH firewall) pin 56 | 520 | ST185-AB-cabin | SGND |
| 6 | w_gnd_a_c44 | 20 | Blue | 🟨🟨 | Gnd flex + MAP | DRB102 receptacle (RH firewall) pin 18 | 520 | ST185-AB-cabin | SGND |
| 7 | w40 | 20 | Yellow | 🟨🟨 | Pedal 1 ground | accelerator pedal sensor pin 5 | 1200 | ST185-AB-cabin | SGND |
| 8 | w_aps_gnd2 | 20 | Yellow | 🟨🟨 | Pedal 2 ground | accelerator pedal sensor pin 2 | 1200 | ST185-AB-cabin | SGND |
| 9 | cab_fspur_gnd | 20 | Yellow | 🟨🟨 | Front VRC ground | to front VR conditioner pin 2 | 640 | ST185-AB-cabin | SGND |
| 10 | | | | | spare, sealing plug | | | | |
| 11 | | | | | spare, sealing plug | | | | |
| 12 | | | | | spare, sealing plug | | | | |

| OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|
| BULKHEAD pin 73 | 1660 | ST185-AB-engine | KNOCK1 |

| OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|
| Sensor screen join - RL + RR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Rear | WSS_RL |

| OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|
| Sensor screen join - RL + RR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Rear | WSS_RR |

| OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|
| BULKHEAD pin 57 | 1100 | ST185-AB-engine | TURBO |

| OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|
| BULKHEAD pin 69 | 1580 | ST185-AB-engine | VSS |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| A | 2026-10-20 | DG | First issue: every harness of the master in one drawing set (replaces the ST185-AB-cabin pilot, R... |

## ST185-harness - every wiring harness of the car, one drawing set

Pinout tables - every pin of every connector, bus, splice and end

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | A | 1 OF 1 | 2159 x 1334 mm | DG | 2026-10-20 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale